

Scale Neglect in LLM Advice: A Source-Backed Diagnostic with Mixed Mitigation Evidence

Anonymous submission

Abstract

LLM advice is increasingly used in settings where the action should change when a decision-relevant magnitude crosses a threshold. We study a narrow failure mode, scale neglect, in which a model acknowledges numbers but fails to alter the final recommendation when the magnitude changes. We build a source-backed diagnostic across LegalBench, BIG-bench arithmetic, and a cognitive-bias decision set, and evaluate five train-free prompting conditions on a 2400-row Gemini 3.5 Flash matrix. The threshold-first protocol is not an unqualified win: it attains macro pair success of 0.870833 versus 0.879167 for direct advice, has slightly higher macro crossover error, and regresses on LegalBench. A separate 480-row source-mapped threshold diagnostic using gpt-5-mini fixes LegalBench support formatting but remains a method failure: LegalBench pair success is 0.65 versus 0.7875 for direct advice. Across 2880 analysis rows, the result is useful as a diagnostic because it exposes saturated cells, scaffold-cue vulnerability, cross-backend comparison limits, and family-specific failure modes that future benchmark repair must address.

Introduction

Advice is different from free-form explanation when a concrete action depends on a magnitude. A recommendation to escalate, approve, abstain, or allocate resources can be correct below one threshold and incorrect above it. The relevant question is not whether a model can restate a number, but whether it changes the final action when the number crosses the governing rule. This paper studies that question under a deliberately narrow diagnostic: source-backed prompts are converted into paired or invariant decision tasks, and the model must produce a structured final action, support, and comparison label.

The problem has a behavioral precedent: people often show scope insensitivity and psychophysical numbing when larger magnitudes should change judgments (Kahneman and Tversky 1979; Carson 1997; Fetherstonhaugh et al. 1998; Bateman et al. 2007). It also has a modern benchmark precedent, since LLM decision-making studies increasingly test uncertainty, sequential choice, and risk-sensitive advice (Liu et al. 2024; Chen, Zhang, and Zhu 2024; Liu et al. 2025; Ouyang, Yun, and Zheng 2024). Cognitive-bias and calibration suites ask whether model outputs reflect human-like deviations or reliable confidence (Echterhoff et al. 2024;

Huang et al. 2026; Chhikara 2025; Wu et al. 2026), while numeracy work shows that language models can mishandle numerical relations (Wallace et al. 2019; Rahman 2025). The gap is that these lines usually do not isolate threshold-crossing magnitude sensitivity in actionable advice.

We therefore ask three concrete questions. First, can a source-backed benchmark detect when an LLM keeps the same action across a threshold crossing? Second, does a train-free threshold-first prompt improve the decision relation rather than just improving explanatory support? Third, do benchmark artifacts themselves contain scaffolding that can make cells saturated and weaken positive claims? Figure 1 summarizes the diagnostic flow.

Our answer is deliberately mixed. The full run contains 2400 scored rows across three families and five train-free conditions. Threshold-first reaches perfect pair success in BIG-bench and Cognitive Bias HF, but it falls to 0.6125 pair success on LegalBench compared with 0.7875 for direct advice. Macro pair success is also slightly lower than direct advice. A separate 480-row source-mapping diagnostic is not a same-backend extension of the full matrix: it uses gpt-5-mini, and therefore supports a failure analysis rather than a cross-model superiority claim. These findings rule out a simple success narrative and motivate a benchmark-hardness supplement rather than a stronger claim.

The paper makes three contributions. First, it defines a paired source-backed diagnostic for scale-sensitive advice, including invariant controls that penalize overreaction to irrelevant magnitude changes. Second, it evaluates direct advice, threshold-first advice, structured calculation, self-consistency thresholding, and calculator-tool thresholding under the same hosted model, scorer, and row schema. Third, it reports calibrated negative evidence: threshold-first is useful as a diagnostic condition, and a later source-mapping ablation fixes one support-formatting failure mode, but neither result supports an unqualified mitigation claim.

Related Work

LLM decision-making work has explored uncertainty, sequential choice, and risk-sensitive behavior. DeLLMa and related sequential-decision methods introduce inference-time procedures for uncertain decisions, while economic-risk studies probe whether model choices align with human preference patterns (Liu et al. 2024; Chen, Zhang, and Zhu

Scale-Neglect Evaluation Flow

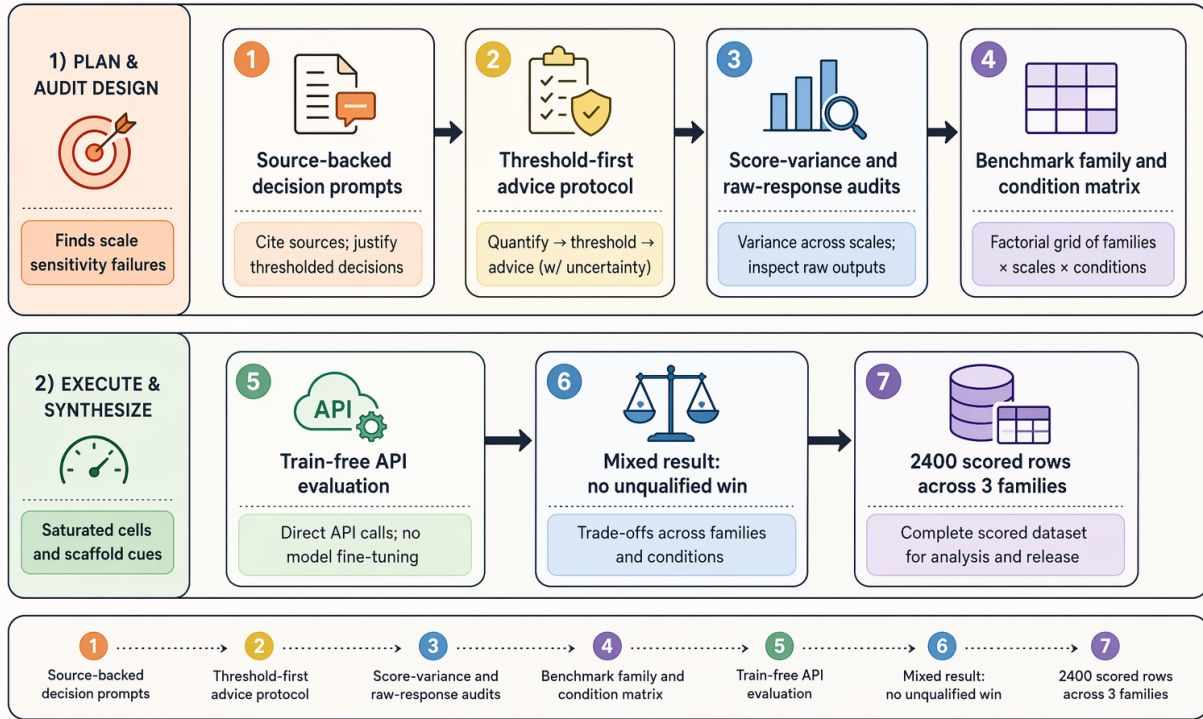


Figure 1: The study’s central workflow is diagnostic rather than promotional: source-backed decision prompts are evaluated under train-free conditions, checked for saturated score patterns, and converted into calibrated claim evidence.

2024; Liu et al. 2025; Ouyang, Yun, and Zheng 2024). Our task differs because the correct decision is controlled by a threshold relation inside paired source-backed prompts, not by long-horizon reward or persona-conditioned risk preference.

Cognitive-bias benchmarks provide the closest methodological precedent. BiasBuster and CogBias organize prompts around human-like deviations and test mitigation strategies (Echterhoff et al. 2024; Huang et al. 2026; Chhikara 2025; Wu et al. 2026). Calibration work similarly asks whether expressed confidence supports decision-level reliability. Those efforts motivate our measurement style, but our benchmark isolates whether the same source item changes action under a magnitude sweep.

The distinction matters because many bias and calibration suites score a single response against a single label. That design can identify biased final choices, but it does not require a model to preserve a source rule across a controlled low-high pair. In a scale-sensitive advice setting, the relation between two responses is part of the target behavior. A model that gives one locally plausible answer below the threshold and another locally plausible answer above it may still fail if the two answers are inconsistent with the same governing rule. This paper therefore treats paired consistency as a first-class metric rather than an optional robustness check.

Numeracy studies show that language models can strug-

gle with number representation and numerical reasoning (Wallace et al. 2019; Rahman 2025). Scale neglect is not identical to arithmetic failure. In many saturated examples, the model states the threshold and comparison correctly. The error appears when correct local support does not translate into a reliable paired action relation, or when prompt scaffolding makes the task too easy to diagnose real difficulty.

The behavioral-decision anchors are also narrower than a generic “LLMs are biased” frame. Scope insensitivity and psychophysical numbing describe cases where larger stakes or quantities do not proportionally change judgment (Carson 1997; Fetherstonhaugh et al. 1998; Bateman et al. 2007). The diagnostic here borrows that concern but translates it into an actionable language-model test: when the source rule says the action should change at a threshold, does the generated advice actually change? That translation is why the benchmark reports action, support, comparison, pair success, crossover, and invariant stability instead of only a single accuracy score.

Method

We construct a benchmark adapter around source-backed items from three families: LegalBench clauses, BIG-bench arithmetic items, and cognitive-bias decision prompts. Each retained source item becomes either an actionable threshold pair or an invariant control. In actionable pairs, the low and

high rows share a source item but cross a declared threshold; a correct system must change the final action and support the comparison. In invariant controls, the magnitude changes while the governing rule says the final action should remain stable.

The evaluated output schema has three paper-facing fields: a final action, a support statement, and a comparison label. The scorer checks action-only accuracy, support-only accuracy, comparison accuracy, pair-level threshold-sensitive success, crossover error, invariant false-change, malformed output, and refusal. Pair-level success is the central diagnostic. It requires both rows of a paired decision to be right and relation-aware.

Action-only accuracy asks whether the final recommendation matches the expected action for each row. Support-only accuracy asks whether the explanation or cited reason is compatible with the governing source rule. Comparison accuracy asks whether the model identifies the threshold relation.

These row-level fields are useful but incomplete. A model can state the right comparison on both rows while giving a final action that fails to change across the pair. Pair-level success therefore aggregates over the paired source identifier and requires the relation-aware outcome, not just a correct local label.

Crossover error captures the specific failure in which a response crosses the threshold relation incorrectly. Invariant false-change captures the complementary failure in which the response changes when the rule says the action should stay stable. Keeping both metrics prevents an intervention from looking good merely by making the model sensitive to every number. A high-sensitivity prompt can lower crossover errors and still be unsafe if it changes invariant decisions too often. This distinction explains the calculator-tool condition in the current evidence.

We evaluate five train-free conditions. Direct advice asks for the recommendation directly. Threshold-first asks the model to identify the quantity and threshold before emitting the action. Structured calculation asks for compact inequality reasoning without the same protocol wording.

The remaining two conditions test whether extra comparison machinery changes the relation metric. Self-consistency samples threshold comparisons and votes inside one completion. Calculator-tool thresholding makes deterministic threshold arithmetic more explicit. We treat all five as hypotheses under test rather than assuming any condition should win.

The saturation audit is part of the method because it changes the interpretation. Three cells are constant-perfect in the full evaluation: BIG-bench threshold-first, Cognitive Bias HF direct advice, and Cognitive Bias HF threshold-first. A follow-up inspection of 624 saturated rows finds exact labels, explicit threshold reasoning, and prompt over-scaffolding rather than substantive variation in difficulty.

That audit rules out stale collection and scoring defects, while leaving prompt over-scaffolding as the most plausible benchmark-design explanation. The paper therefore treats saturated cells as evidence about the diagnostic, not as evidence that the underlying advice task is solved.

The transformation is intentionally simple enough to audit. For each actionable source item, the benchmark stores a pair identifier, the source family, a low or high magnitude, the expected comparison relation, the expected action, and the governing rule. The paired design prevents a response from being judged only row by row.

A model that always says to take action can look strong on one side of a distribution, but it fails the pair relation when the action should flip. Conversely, a model that states the right threshold while preserving the wrong action receives support credit but not pair-level success. This separation matters because the analysis finds exactly this kind of metric disagreement.

The invariant controls serve a second role. They detect whether a threshold-focused instruction causes the model to overreact to any number change. This matters for a train-free intervention because a prompt can improve sensitivity by making every number look decisive, which would be harmful in tasks where the rule says the action should remain stable. We therefore treat invariant false-change as a constraint on any claimed mitigation. Calculator-tool thresholding illustrates the tradeoff: it has zero macro crossover error, but its invariant false-change is higher than the planned tolerance.

We also keep response validation tied to the structured output schema. The scorer first trusts exact structured fields, then uses conservative fallback checks only when needed. The saturated-cell inspection does not indicate that the perfect cells come from scoring defects; instead, it points to benchmark scaffolding. That result changes the paper claim: the benchmark is useful enough to reveal family-specific behavior, but some cells are too scaffolded to support a general claim about robust magnitude reasoning.

Experimental Setup

The full evaluation contains 2400 scored model responses. It covers 480 source-backed task rows crossed with five train-free conditions, yielding 160 scored rows per family-condition cell. The evaluated hosted model is Gemini 3.5 Flash through an OpenAI-compatible chat-completions interface. The study uses inference-time prompting rather than model training.

After the full matrix exposed a LegalBench regression, we ran one bounded source-mapped threshold diagnostic on the same 480 source-backed rows. This ablation used gpt-5-mini and is reported separately from the Gemini matrix. It evaluates one hosted completion per row with a 180-token cap and provider-default temperature. Its purpose is failure diagnosis, not cross-backend ranking: if it improves one metric while losing to the full-run direct-advice baseline on LegalBench, the result is evidence about the failure mode rather than evidence that a new backend or method is superior.

Each standard condition uses one hosted completion per row with a 120-token cap. Temperature and top- p were not explicitly set in the evaluation request, so both are reported as provider defaults rather than reconstructed after the fact. The self-consistency condition is not five separate API calls: it asks one completion to simulate five independent

threshold-comparison votes and emit the voted final action under the same token cap.

The primary metrics are pair-level threshold-sensitive success, crossover error, and invariant false-change. Secondary metrics are action-only accuracy, support-only accuracy, comparison accuracy, malformed rate, refusal rate, and token usage. Confidence intervals are paired bootstrap intervals over family-condition pair/control identifiers, and matched pair-level comparisons use exact McNemar tests with Holm correction.

The setup intentionally keeps negative evidence. A failed mitigation claim is not discarded, and saturated cells are not treated as proof of model competence. The benchmark analysis shows that these cells need a harder adversarial split, hard-negative set, and scaffold-cue audit before stronger claims can be made.

The source families play different roles. LegalBench supplies policy-like legal decision items where the final action depends on both the original source label and a magnitude such as affected claimants. BIG-bench arithmetic supplies numerical transformations where the threshold comparison is more explicit. Cognitive Bias HF supplies decision prompts close to the stated bias theme. The three-family mix is valuable because a single aggregate would hide the LegalBench regression behind easier cells.

The benchmark adapter preserves this family distinction throughout scoring. Each family contributes the same number of scored rows per condition, so the macro metrics are not dominated by one source. At the same time, the families are not interchangeable. BIG-bench is closer to explicit numerical comparison, LegalBench combines source labels with threshold magnitudes, and Cognitive Bias HF is closest to the paper’s motivating bias literature. The setup therefore treats family-stratified results as necessary evidence rather than secondary subgroup analysis.

The benchmark rows are also deliberately paired. A paired threshold item creates two related prompts around the same source-backed rule, and the expected final action changes only when the governing threshold relation changes. An invariant control changes a magnitude while holding the correct action stable. This design gives the scorer two ways to reject a shallow response: it can fail to flip when the threshold is crossed, or it can flip when the rule says the action should remain unchanged. Those two errors are reported separately as crossover error and invariant false-change.

The five train-free conditions are all evaluated under the same scorer and row schema. Direct advice asks for a decision without the threshold-first decomposition. Threshold-first asks for the quantity, rule, comparison, and final action. Structured calculation asks for compact calculation or inequality reasoning without using the same protocol wording. Self-consistency thresholding samples threshold comparisons and votes. Calculator-tool thresholding is an upper-bound style prompt where deterministic threshold arithmetic is made more explicit. These conditions are baselines for interpretation, not evidence of training or model adaptation.

The train-free restriction is part of the experimental question. The study does not fine-tune, distill, retrieve ex-

Table 1: Macro metrics: threshold-first trails direct advice on pair success (0.870833 vs. 0.879167).

Condition	Pair	Cross.	Inv. false
Direct advice	0.879167	0.100000	0.016667
Threshold-first	0.870833	0.108333	0.020833
Structured calc.	0.737500	0.075000	0.020833
Self-consistency	0.020833	0.479167	0.070833
Calculator-tool	0.854167	0.000000	0.066667

Note. Pair=pair success, Cross.=crossover error, Inv.=invariant false-change. Full row-level action, support, and comparison metrics are in the generated results artifact.

tra task-specific examples, or change the source benchmark after seeing failures. The evidence therefore tests whether prompting and scoring alone expose or improve scale-sensitive advice. When self-consistency fails or LegalBench regresses, the failure is not hidden behind a training recipe or a moving benchmark target.

The row count follows directly from this crossing design. The full run has 480 source-backed task rows per condition, distributed as 160 rows per benchmark family. Crossing those rows with five train-free conditions yields 2400 scored rows. Family-condition cells are balanced at the row level, while paired metrics are computed over the corresponding pair/control identifiers.

The statistical protocol is also conservative. Bootstrap intervals are paired over source pair/control identifiers, not over arbitrary response rows. McNemar comparisons use matched pair identifiers within each source family. The paper uses those tests to identify where the result is directional and where it is saturated; it does not use them to claim broad generalization beyond this benchmark slice.

The cost, malformed, and refusal fields are retained for interpretation but are not the primary objective. A prompt that adds tokens can be reasonable if it improves the relation metric and keeps invariant controls stable. A prompt that improves only row-level support is not a useful mitigation. In the current run, output-format failures do not explain the saturated cells or LegalBench regression.

Results

Table 1 gives the macro result. Threshold-first leads action and comparison accuracy, but its pair success is 0.870833 versus 0.879167 for direct advice, with slightly worse crossover error. Table 2 uses 80 tasks / 160 rows per family, and Figure 2 explains the aggregate by family.

Analysis / Ablation

Metric disagreement. Local metric gains do not reliably transfer to paired decisions. Structured calculation lowers crossover while losing pair success; calculator-tool removes crossover but raises invariant false changes. Self-consistency is the negative control.

Table 3: Source mapping fixes LegalBench support (1.0) but not pair success (0.65).

Table 2: Family matrix: threshold-first improves BIG-bench but trails direct advice on LegalBench (0.6125 vs. 0.7875).

Benchmark/source	Backend	Method	Rows	Budget	Pair	Cross.	Inv.
BIG-bench	Gemini 3.5 Flash	Direct	160	1 completion	0.8500	0.1500	0.0000
BIG-bench	Gemini 3.5 Flash	Threshold-first	160	1 completion	1.0000	0.0000	0.0000
BIG-bench	Gemini 3.5 Flash	Structured calc.	160	1 completion	0.5625	0.2250	0.0000
BIG-bench	Gemini 3.5 Flash	Self-cons.	160	5 simulated votes	0.0125	0.4875	0.0000
BIG-bench	Gemini 3.5 Flash	Calc-tool	160	calc hint	0.9750	0.0000	0.0000
Cognitive Bias HF	Gemini 3.5 Flash	Direct	160	1 completion	1.0000	0.0000	0.0000
Cognitive Bias HF	Gemini 3.5 Flash	Threshold-first	160	1 completion	1.0000	0.0000	0.0000
Cognitive Bias HF	Gemini 3.5 Flash	Structured calc.	160	1 completion	0.9625	0.0000	0.0000
Cognitive Bias HF	Gemini 3.5 Flash	Self-cons.	160	5 simulated votes	0.0375	0.4625	0.0000
Cognitive Bias HF	Gemini 3.5 Flash	Calc-tool	160	calc hint	0.9125	0.0000	0.0000
LegalBench	Gemini 3.5 Flash	Direct	160	1 completion	0.7875	0.1500	0.0500
LegalBench	Gemini 3.5 Flash	Threshold-first	160	1 completion	0.6125	0.3250	0.0625
LegalBench	Gemini 3.5 Flash	Structured calc.	160	1 completion	0.6875	0.0000	0.0625
LegalBench	Gemini 3.5 Flash	Self-cons.	160	5 simulated votes	0.0125	0.4875	0.2125
LegalBench	Gemini 3.5 Flash	Calc-tool	160	calc hint	0.6750	0.0000	0.2000

Note. Pair=pair success, Cross.=crossover error, Inv.=invariant false-change. Each row uses 160 scored responses from 80 pair/control identifiers.

Family	Pair	Support	Interpretation
BIG-bench	1.0000	1.0000	saturated
Cognitive Bias HF	1.0000	1.0000	saturated
LegalBench	0.6500	1.0000	method failure

Diagnostic only. gpt-5-mini, 480 rows, one hosted completion, 180-token cap, provider-default temperature; not a same-backend Gemini extension.

Source-mapping diagnostic. Table 3 reports the ablation. LegalBench support reaches 1.0, but pair success is 0.65 vs. direct advice at 0.7875. The run is diagnostic, not a same-backend win.

Family evidence. The family signs split; the planned crossover-reduction claim is unsupported. LegalBench is the stress test. Direct advice reaches 0.7875 pair success, while threshold-first reaches 0.6125. BIG-bench gives the opposite local story: threshold-first is perfect where the comparison is explicit. Cognitive Bias HF is saturated because direct advice and threshold-first both reach perfect pair success. The positive claim is local, not general.

Statistical interpretation. In BIG-bench, threshold-first has 12 unique pair successes (Holm-adjusted McNemar $p = 0.0048828125$). In LegalBench, the sign reverses: threshold-first has 11 unique pair successes, while direct advice has 25 on the same 80 comparable pair identifiers. The unadjusted LegalBench test is negative for threshold-first, and the Holm-adjusted value does not support a broad family-level claim.

Failure Cases

Source-mapping method failure. The LegalBench source-mapping audit isolates 48 failed rows. Support and comparison are correct for all of them. The failures are action-level: 40 invariant-control action failures and

Table 4: LegalBench failure decomposition: source mapping removes crossover and support-pair failures but leaves 28 action-wrong pairs.

Method	Pair	Cross.	Inv.	Action	Support
Direct	63	12	4	4	13
Threshold	49	26	5	5	26
Structured	55	0	5	5	21
Calc-tool	54	0	16	24	23
Self-cons.	1	39	17	35	79
Source-map	52	0	8	28	0

Note. Counts are matched LegalBench pair identifiers. Pair=pair successes; Cross.=crossover-error pairs; Inv.=invariant false-change pairs; Action/Support=count of pairs with that failure component.

8 invariant false changes. The ablation fixed support formatting, but the final action relation remained wrong. This is LegalBench method failure, not support for a stronger mitigation claim.

LegalBench regression. The main failure case is LegalBench. Threshold-first can over-index on an explicit threshold relation. It can still mishandle source-specific legal conditions or invariant controls. The result is lower pair success than direct advice even when the comparison label is often correct.

Failure taxonomy. Table 4 makes the diagnosis concrete. On LegalBench, threshold-first records 49 pair successes while direct advice records 63. Support-failure pairs double from 13 to 26, and crossover-error pairs rise from 12 to 26.

The source-mapping diagnostic changes the error profile without clearing the family. It removes crossover and

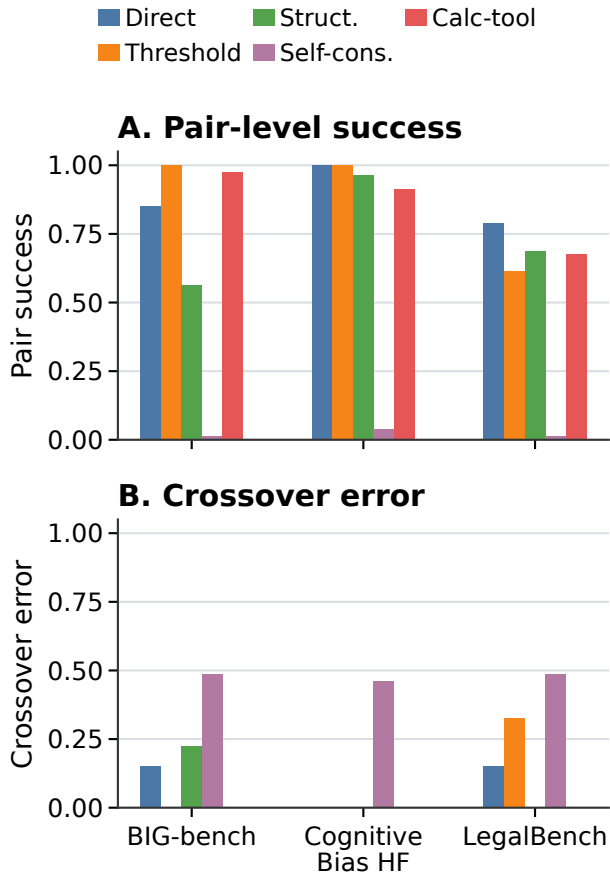


Figure 2: Family-level results show the core mixed pattern: threshold-first is strongest on BIG-bench, tied in saturated Cognitive Bias HF cells, and worse than direct advice on LegalBench.

support-pair failures, but it leaves 28 action-wrong pairs and 8 invariant false-change pairs. The local comparison step is therefore not enough. The model can name a relation and still fail to preserve the source-conditioned action mapping across the pair.

Paired source relations. This failure is not visible from the easiest row-level fields. Threshold-first can still look organized: it usually emits the comparison label and often names the threshold.

The damage appears when the two rows tied to the same source item are judged together. In LegalBench, the direct condition keeps more paired source relations intact. Threshold-first shifts attention toward the explicit numerical threshold and away from the legal condition that determines the action.

Saturation and metric masking. The second failure case is benchmark saturation. The raw audit shows 624 saturated rows with exact action and comparison labels and explicit threshold or invariant reasoning.

Because all audited saturated rows also show prompt

over-scaffolding, the correct response is not to claim robust competence. The correct response is to create a harder benchmark split with adversarial prompts, hard negatives, and scaffold-cue checks.

Together, the LegalBench regression and the saturation audit explain why the paper treats LegalBench as the stress test rather than as a noisy subgroup.

The saturation audit also narrows what the paper can and cannot conclude. It rules out missing raw outputs, rescore mismatches, parser leniency, and gold-label ambiguity as explanations for the constant-perfect cells. It does not prove robust scale-sensitive reasoning. Instead, the perfect cells are benchmark-design evidence: some source transformations are too transparent under the current scaffold, while LegalBench remains the family where the source condition and threshold relation must both be preserved.

The third failure case is metric masking. A condition can look strong on action-only accuracy while failing pair-level relation checks, or can remove crossover errors while creating invariant false changes. Calculator-tool thresholding illustrates the tradeoff: it removes macro crossover errors, but its invariant false-change rate reaches 0.066667. Self-consistency gives a second warning: more intermediate reasoning text does not imply better scale-sensitive advice when the paired relation collapses.

Use of the result. These failure cases shape the paper’s intended use. The study is not a deployment recommendation for threshold-first prompting; it is a diagnostic benchmark showing where paired tasks, invariant controls, family reports, and joint evaluation of support statements and final actions are necessary. More explicit reasoning is insufficient unless it also preserves the paired source relation and invariant-control stability.

Claims ruled out. The current evidence rules out three stronger claims. First, threshold-first is not a robust mitigation because its macro pair success trails direct advice and its macro crossover error is higher. Second, the cognitive-bias family is not solved, because two relevant cells are saturated and over-scaffolded. Third, calculator-style prompting is not a robust upper bound, because low crossover error comes with too many false changes on invariant controls.

Those ruled-out claims define the reporting standard. Table 1, Table 2, Table 3, and Figure 2 must be read together: the macro table shows the support/action mismatch, the family matrix exposes the LegalBench reversal, the diagnostic table separates the gpt-5-mini source-mapping ablation, and the figure makes saturated cells visible.

For benchmark repair, the target is a harder source-backed split rather than a larger copy of the same prompt patterns. Near-threshold cases should reduce the numerical margin between low and high rows. Hard negatives should preserve the source rule while varying irrelevant magnitudes. Cue checks should test whether final labels can be inferred from prompt form alone. The retained contribution is that the measurement stack separates three states: a real improvement in the pair relation, a row-level improvement that does not survive pairing, and a benchmark cell whose perfect

score is too saturated to separate competence from scaffolding.

Future methods should be judged by a conjunctive standard: improve pair success, reduce crossover errors, avoid raising invariant false changes within each family, and keep support/action fields visible. Without that full report, a method could appear successful by selecting only the metric or family where its prompt style is favored.

Conclusion

This paper reports a diagnostic result. Threshold-first prompting can expose scale-sensitive reasoning failures, but the current full evaluation does not support an unqualified mitigation claim.

The strongest supported claim is about measurement. Across 2400 scored rows, the benchmark separates action, support, comparison, pair success, crossover, and invariant false changes. Threshold-first improves some local fields, but macro pair success remains below direct advice. Crossover error is slightly higher, and LegalBench regresses.

This result changes how scale-sensitive advice should be evaluated. A row can look correct in isolation while the paired source relation is wrong, and a family can look solved while the prompt scaffold is too transparent to support a robust reasoning claim. The benchmark is useful because it makes those cases visible under one protocol: the same source families, row counts, completion budget, and scorer expose both the favorable BIG-bench cell and the unfavorable LegalBench cell. The separate 480-row source-mapping diagnostic reinforces the same boundary: support formatting can be fixed while LegalBench pair success remains below direct advice.

The practical contribution is therefore a measurement contract. Reports should identify the evaluated backend, task split, decoding budget, and paired metric alongside the family-level scores. They should also keep saturated cells and invariant-control errors in the main evidence rather than treating them as appendix-only diagnostics. Without those fields, a later method could appear to improve scale-sensitive advice by optimizing a local label while leaving the source-conditioned decision relation broken.

The acceptance standard for a stronger mitigation paper is therefore stricter: beat direct advice on pair success within each family, reduce crossover errors without raising invariant false changes, and survive a harder split with fewer scaffold cues. LegalBench and saturated cells show why all three tests are necessary.

The negative cells are part of the contribution. They show that a prompt can make the threshold comparison explicit while still losing the source-conditioned action relation. They also show that perfect cells need saturation checks before they can be treated as reasoning evidence. A useful follow-up should therefore report the same family matrix and error decomposition, even if it introduces a stronger prompting or tool-use condition.

Future evaluations should preserve paired accounting and saturated-cell flags while adding near-boundary examples, hard negatives, and cue checks. The immediate benchmark

repair is not to repeat the same transparent rows at larger scale. It is to make the source rule harder to shortcut: reduce the numerical margin around thresholds, add irrelevant but salient quantities to invariant controls, and test whether a shallow response can infer the final label from prompt form alone. A future mitigation should also be judged under a same-backend protocol, because the source-mapping diagnostic in this paper uses gpt-5-mini while the main full matrix uses Gemini 3.5 Flash.

This conclusion keeps the paper’s claim boundary narrow. The evidence supports a source-backed diagnostic, a mixed threshold-first result, a LegalBench regression, and a measured source-mapping method failure. It does not support a deployment recommendation, a model-superiority claim, or a statement that scale-sensitive advice is solved in saturated families. That boundary is useful: it tells later work exactly which evidence must change before a stronger claim is warranted.

The practical reporting lesson is also concrete. A future paper should not report only a macro accuracy or a single family win for scale-sensitive advice. It should show the family matrix, the paired relation metric, the invariant-control error, and the source-mapping or support-field audit when an intervention changes explanations. In the present evidence, those views disagree in informative ways: threshold-first has strong row-level action and comparison accuracy, direct advice is stronger on LegalBench pair success, calculator-tool thresholding removes crossover errors while raising invariant false changes, and source mapping repairs LegalBench support while leaving the action relation broken. These disagreements are the diagnostic value of the benchmark.

The same discipline should apply to future source-mapping variants. A successful variant would need to be evaluated on the same backend as the full matrix or under a deliberately balanced cross-backend protocol; otherwise, a change in model can be confused with a change in method. It would also need to clear LegalBench without relying on the saturated BigBench and Cognitive Bias HF cells. The current source-mapping ablation is therefore useful because it localizes the remaining failure: support evidence can be formatted correctly, yet the final source-conditioned action can still be wrong. That is a sharper target for the next benchmark and method iteration than the original aggregate score alone.

The study also clarifies what should be preserved when the benchmark is revised. The row schema should keep final action, support, comparison, malformed, refusal, and paired relation fields separate. The family reports should remain visible in the main paper because the aggregate hides the LegalBench reversal. The failed-row audit should remain part of the analysis whenever an intervention changes support formatting, because a support repair can mask a final-action failure. These are simple reporting requirements, but they prevent the main failure observed here: a locally plausible threshold explanation being mistaken for a correct source-conditioned recommendation.

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Reproducibility Checklist

Scope. The paper is based on one completed full evaluation with 2400 scored rows, three benchmark families, and five train-free conditions, plus a separate 480-row source-mapping diagnostic ablation reported only as failure analysis.

Artifacts. The reproducibility bundle includes scored model responses, generated result tables, statistical comparisons, benchmark-family summaries, the LegalBench source-mapping failed-row audit, and the source needed to rebuild the manuscript tables and plots.

Limitations. The paper explicitly preserves saturated-cell caveats, the LegalBench regression, and the gpt-5-mini cross-model caveat for the source-mapping diagnostic.

Appendix: Artifact Provenance

The supplementary package contains the completed scored responses, family-condition metric tables, paired statistical comparisons, saturated-cell notes, and figure metadata. The main text uses these materials to rebuild the macro table, the cross-benchmark matrix, and the two family-level plots. This appendix describes artifact categories only; package-level identifiers are kept outside the rendered manuscript.